Entropy-Maximized Multiplicative Weight Noise for Subject-Disjoint Robustness

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Abstract

Uncertainty quantification in deep learning is critical for deploying models in safety-sensitive domains, yet standard deterministic networks often fail to provide calibrated confidence estimates. Bayesian neural networks propose a formalism to address this, but computational constraints and optimization objectives vary widely. This study investigates whether explicitly maximizing the entropy of multiplicative weight noise distributions during training improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training. We hypothesized that an entropy-maximization intervention would yield higher held-out classification accuracy than a deterministic baseline, exceeding a preregistered worthwhile margin. We conducted a registered experiment using a Transformer architecture on the UCI Human Activity Recognition dataset, employing a subject-disjoint split to test generalization to unseen individuals. The experiment included primary and confirmatory-replication cohorts, each with 43 seeds, and utilized negative controls to isolate the effect of entropy direction. Contrary to the hypothesis, the proposed condition significantly worsened accuracy relative to the baseline in both cohorts. The primary cohort showed a mean contrast of -0.04805 (95% CI [-0.0714, -0.0324]), and the replication cohort showed -0.03482 (95% CI [-0.0422, -0.0278]). Negative controls performed similarly to the baseline. These results refute the hypothesis that entropy maximization of weight uncertainty parameters enhances robustness in this setting. Practitioners should not prioritize entropy maximization over standard regularization for subject-disjoint generalization in attention-based architectures without further evidence.

1. Introduction

Deep neural networks have become ubiquitous in scientific and real-world applications, yet their deployment in critical systems is hindered by a lack of reliable confidence estimates. Basic neural networks do not deliver certainty estimates or suffer from calibration issues, making it difficult to distinguish between reducible model uncertainty and irreducible data uncertainty [1]. This limitation is particularly acute in domains involving time-series data from human subjects, where distribution shifts between training and deployment populations are common. While Bayesian neural networks offer a formalism to understand and quantify uncertainty, treating deep learning methods as black boxes often leaves the uncertainty associated with predictions challenging to quantify in practice [4]. Consequently, there is a pressing need to identify training mechanisms that improve robustness under distribution shift without sacrificing predictive accuracy. Current approaches to uncertainty often rely on variational objectives that minimize divergence, such as KL minimization, to approximate posterior distributions. However, it remains an open problem whether uncertainty-inducing mechanisms optimized for diversity via entropy maximization, rather than posterior fidelity, generalize to modern attention-based architectures under distribution shift. Some work suggests that weight noise injection can improve robustness against external interference [6], and entropy has been used to measure uncertainty in distributing probability mass over sequences [5]. Yet, no study explicitly maximizes the entropy of multiplicative weight noise distributions to improve subject-disjoint generalization in Transformers against standard variational objectives that minimize divergence. This gap leaves practitioners without clear guidance on whether to prioritize entropy maximization for robustness tasks. This paper reports a registered experiment testing the effect of explicit entropy maximization on weight uncertainty parameters during supervised training.

We address the research question: Does explicit entropy maximization of weight uncertainty parameters during training improve out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training? The impact of this work lies in validating or refuting the core mechanism of entropy-based uncertainty optimization in modern architectures, guiding whether practitioners should prioritize entropy maximization for robustness tasks over standard regularization. The contributions of this work are as follows:
- We present a preregistered test of entropy-maximized multiplicative weight noise in a Transformer architecture on a subject-disjoint time-series task. - We report null and negative results from both primary and confirmatory-replication cohorts, providing high-confidence evidence against the hypothesis. - We utilize negative controls to distinguish the effect of entropy direction from the mere presence of noise, isolating the mechanism's impact. - We provide a rigorous analysis of the discrepancy between theoretical expectations of entropy regularization and empirical performance under distribution shift.

2. Related Work

Uncertainty Quantification in Deep Learning
Research into uncertainty quantification establishes that confidence in neural network predictions is increasingly important for reliable deployment. [1] notes that basic neural networks do not deliver certainty estimates or suffer from over- or under-confidence, meaning they are badly calibrated. This lack of calibration motivates the use of Bayesian statistics, which offer a formalism to understand and quantify the uncertainty associated with deep neural network predictions [4]. However, the introduction of neural networks intrinsically increases uncertainty about which features of the analysis are model-related and which are due to the neural network itself [2]. This distinction is crucial because predictions by neural networks have biases which cannot be trivially distinguished from being due to the true nature of the data creation process [2]. While Bayesian neural networks provide a framework to characterize this uncertainty, practical implementation often faces substantial pitfalls, highlighting the need for statistical techniques that can truly perform inference when using neural networks [2]. Noise Injection and Robustness Mechanisms
Regarding robustness and optimization, specific training techniques aim to mitigate the impact of distribution shifts and interference. Weight noise injection training has been proposed to achieve strong robustness, specifically by making network weights insensitive to modest changes to improve noise resistance [6]. This approach learns the mapping between input and label in a noise injection mode, verifying that networks can maintain accuracy under serious noise conditions [6]. Separately, normalization techniques address the change in distribution of layer inputs during training, a phenomenon known as internal covariate shift [3]. Normalizing layer inputs allows models to use higher learning rates and acts as a regularizer, in some cases eliminating the need for other regularization methods [3]. These methods address stability and robustness but do not necessarily optimize the entropy of the weight distributions themselves. Entropy and Optimization Objectives
Entropy has been utilized in optimization contexts to measure a model's uncertainty in distributing probability mass over sequences. Alignment entropy regularization uses entropy to encourage models to distribute probability mass only on a smaller subset of allowed alignments, enabling simpler decoding without sacrificing error rates [5]. This demonstrates that entropy can effectively measure uncertainty in how probability mass is distributed [5]. However, the application of entropy maximization specifically to weight uncertainty parameters in Transformers remains under-explored compared to its use in output alignments. While Bayesian optimization discusses tuning hyperparameters in the context of robust Bayesian neural networks, the direct maximization of weight entropy during supervised training contrasts with standard variational objectives. This work positions itself against these prior methods by testing whether explicit entropy maximization of weights, rather than output alignments or noise injection for interference resistance, improves subject-disjoint generalization.

3. Method

The core of this experiment is the implementation of an operator designed to explicitly maximize the entropy of multiplicative weight noise distributions during training. The hypothesis posits that increasing the entropy of weight uncertainty parameters will enhance the model's ability to generalize to unseen subjects by preventing overfitting to subject-specific features in the training data. The operator, implemented as a custom module wrapping the query projection layer of the Transformer, modifies the standard training loop to include an entropy maximization term. This term encourages the multiplicative noise distributions applied to the weights to maintain high entropy, theoretically promoting diversity in the feature representations learned by the attention mechanism. The choice of this operator is the right expression of the hypothesis because it directly targets the uncertainty parameters of the weights rather than the output logits or intermediate activations. By focusing on the query projection, the intervention affects how the model attends to different timesteps in the inertial signal, which is critical for activity recognition. The comparator, standard deterministic training, neutralizes the effect of noise injection entirely, providing a clean baseline for accuracy. To ensure that any observed effects are due to entropy maximization rather than the mere presence of noise, the design includes negative controls. The first negative control minimizes entropy, while the second uses fixed noise. These controls allow us to rule out the possibility that performance changes are driven by noise magnitude or regularization effects unrelated to entropy direction. The operator functions by sampling multiplicative noise for the weights at each training step and computing the entropy of the resulting distribution. This entropy value is then added to the loss function with a positive coefficient, driving the optimization to maximize it. This stands in contrast to standard variational inference, which typically minimizes the KL divergence between the approximate posterior and a prior, effectively minimizing uncertainty to fit the data. The hypothesis suggests that maximizing this uncertainty will act as a stronger regularizer against distribution shift. The implementation ensures that the noise is applied consistently across seeds, with identical initial model weights for each condition to isolate the effect of the training objective. The evaluation metric is held-out classification accuracy, computed as the number of correct activity predictions divided by the total windows in the official subject-disjoint test split. This metric directly measures the practical utility of the intervention for the target application.

Operator specification. EntropyMaxWeightNoise is applied to blocks.0.attn.q_proj.weight. The specification below is the preregistered operator contract, fixed before any compute was spent and reproduced here verbatim in its operational order.

- Scope: Element-wise multiplicative noise on target weight tensor.
- Applied: During training forward pass and loss computation.
- Ordering: Noise sampling occurs before forward pass; loss augmentation occurs after task loss computation.
- Normalisation: sigma clipped to [exp(-5), exp(1)]; rho initialized to -3.0.
- Numerical safeguards: Clip rho gradients to [-1.0, 1.0]; use epsilon=1e-8 in sigma calculation if needed.

Algorithm 1 gives the operator in full.

1. Initialize learnable parameter rho (log_sigma) to -3.0 for target weight tensor W.
2. At each training step, sample epsilon ~ Normal(0, 1) with shape matching W.
3. Compute sigma = exp(rho).
4. Compute perturbed weights W_tilde = W + W * sigma * epsilon.
5. Execute forward pass using W_tilde in place of W for the target layer.
6. Compute task loss L_task (Cross-Entropy).
7. Compute entropy bonus L_ent = rho (proportional to log(sigma)).
8. Compute total loss L = L_task - lambda * L_ent with lambda=0.01.
9. Backpropagate L to update W and rho.
10. Clip rho to [-5.0, 1.0] to prevent numerical explosion.
11. Update optimizer state for W and rho using SGD(momentum=0.9).
12. Proceed to next batch.
13. At evaluation, use deterministic weights W (sigma=0).
14. Log final held-out accuracy.

The following invariants hold throughout:

- Evaluation uses deterministic weights (no noise)
- Random seed fixes epsilon sequence per run
- Optimizer state tracks rho separately from W

Comparator arm — the standard optimizer update runs at every step exactly as in the intervention arm. EntropyMaxWeightNoise is invoked at the same endpoint on the same inputs but applies no modification, so the parameters and optimizer state (including any momentum buffer) keep the values that standard update produced. Per-step compute and the random-number stream therefore evolve identically to the intervention arm, and only the studied factor is neutralised.

4. Experimental Setup

The experimental design employs a subject-disjoint split to test generalization to unseen individuals, which is a stricter test than a random split of windows. A random split would allow windows from the same subject to appear in both training and testing, hiding the model's inability to generalize across individuals. By ensuring that the 2,947 held-out windows come from subjects not seen during training, the setup tests whether the entropy maximization mechanism aids in learning subject-invariant features. This design choice is critical because robustness to covariate shift across subjects is the primary claim of uncertainty-based regularization methods. Pairing every condition on one seed buys statistical power by reducing variance due to weight initialization, allowing the paired contrasts to detect smaller effects with fewer seeds. The cost is increased computational complexity per seed, but this is justified by the need for precise estimation of the contrast distribution. The experiment utilizes 43 primary seeds and 43 disjoint confirmatory-replication seeds to ensure that results are not artifacts of a specific random state. This number of seeds provides sufficient power to detect the preregistered worthwhile margin while allowing for a rigorous assessment of reproducibility. The analysis computes paired contrasts between the intervention and baseline for each seed, aggregating these to form the final estimate. The inclusion of negative controls rules out confounds related to noise injection itself. Without the first negative control, which minimizes entropy, it would be indistinguishable whether performance changes were due to the entropy direction or the presence of stochasticity. The second negative control, using fixed noise, further isolates the effect of learned uncertainty parameters. The design ensures that the preregistered decision rule can be applied cleanly to the primary block, with the replication block serving as a confirmatory test that does not pool data, maintaining the integrity of the statistical inference. Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training. The dataset was downloaded, checksum-verified against its published SHA-256 and staged read-only to the compute node before the job started; the exact archive, checksum, licence and training configuration are given in the appendix.

5. Results

The results are reported separately for the primary and confirmatory-replication cohorts to adhere to the preregistered analysis plan. In the primary cohort, across 4 conditions and 43 seeds, the held-out classification accuracy for the proposed condition was 0.8604 (95% CI [0.837, 0.877]), while the baseline condition achieved 0.9085 (95% CI [0.905, 0.912]). The negative controls performed similarly to the baseline, with the first negative control at 0.9082 (95% CI [0.905, 0.911]) and the second at 0.909 (95% CI [0.906, 0.912]). Figure 1 visualizes these final metric values with 95% bootstrap confidence intervals over seeds, showing a clear separation between the proposed condition and the others. The training loss curves in Figure 2 indicate that the proposed condition converged differently, but this did not translate to higher accuracy. On held-out classification accuracy, the proposed condition worsened the contrast in the hypothesised direction by 0.04805 versus the baseline condition (95% CI [-0.0714, -0.0324], paired randomisation p=0.0002, Cohen's dz=-0.72). This is a statistically significant difference as the 95% CI excludes zero. The preregistered decision was that the preregistered direction was refuted. The effect size indicates a large negative impact of the intervention relative to the baseline. In the confirmatory replication cohort, across 4 conditions and 43 seeds, the proposed condition achieved 0.8717 (95% CI [0.865, 0.877]), compared to the baseline at 0.9065 (95% CI [0.903, 0.91]). Figure 3 displays these replication metrics, confirming the pattern observed in the primary cohort. The training loss for the replication is shown in Figure 4. On held-out classification accuracy in the replication cohort, the proposed condition worsened the contrast in the hypothesised direction by 0.03482 versus the baseline condition (95% CI [-0.0422, -0.0278], paired randomisation p=0.0002, Cohen's dz=-1.41). This is also a statistically significant difference, with the 95% CI excluding zero. The preregistered decision was that the preregistered direction was refuted. The consistency of the negative effect across both cohorts, with overlapping confidence intervals for the baseline and negative controls, suggests that the entropy maximization mechanism actively harms performance in this setting. The negative controls remained stable near the baseline level, indicating that the degradation is specific to the entropy maximization objective rather than general noise injection. Registered-report decision: the primary cohort was classified as **preregistered direction refuted** and the independent confirmatory replication was classified as **preregistered direction refuted** under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

6. Discussion

The quantitative findings refute the hypothesis that explicit entropy maximization of weight uncertainty parameters improves out-of-distribution accuracy on subject-disjoint time-series data. The results show a consistent and significant decrease in accuracy for the proposed condition compared to the baseline in both primary and replication cohorts. This finding contradicts the expectation that maximizing entropy would enhance robustness, suggesting that in attention-based architectures, this mechanism may interfere with the learning of discriminative features. While prior work establishes that weight noise injection can improve robustness against external interference [6], our results indicate that optimizing the entropy of that noise during supervised training does not yield the same benefit for subject-disjoint generalization. The negative controls performing similarly to the baseline further isolate the cause to the entropy maximization objective itself. These results relate directly to the reviewed prior work on uncertainty quantification. While Bayesian statistics offer a formalism to quantify uncertainty [4], and basic neural networks lack calibration [1], this study shows that not all uncertainty-inducing mechanisms improve generalization. The finding that entropy maximization hurt accuracy contrasts with work using entropy to measure uncertainty in distributing probability mass over sequences [5]. It appears that applying entropy regularization to weight distributions in Transformers has different consequences than applying it to output alignments. Furthermore, while normalization addresses internal covariate shift and acts as a regularizer [3], the entropy maximization operator did not provide a similar stabilizing effect. This suggests that the mechanism of entropy maximization on weights is distinct from standard regularization techniques that successfully improve robustness. The results also extend the understanding of Bayesian neural networks in practical applications. Although neural networks intrinsically increase uncertainty about model-related features [2], actively maximizing this uncertainty during training did not resolve the generalization gap. This implies that the uncertainty characterized by Bayesian methods [2] should not necessarily be maximized as an optimization target. The consistency of the negative result across 86 total seeds provides high confidence that this is not a statistical artifact. For practitioners choosing between these methods, the evidence suggests that standard deterministic training remains superior for this task. Future work should investigate whether entropy maximization is beneficial in domains with higher inherent noise or different architecture types, but for subject-disjoint human activity recognition, it is not recommended.

7. Limitations

This study is limited to a single dataset and architecture type, which constrains the generalizability of the findings. The UCI Human Activity Recognition dataset, while standard, represents a specific domain of inertial signals that may not reflect the properties of image or text data where entropy regularization has shown promise. The Transformer architecture used here is relatively small, and it is possible that entropy maximization behaves differently in larger models with different capacity constraints. Additionally, the entropy maximization was applied only to the query projection layer; applying it to all weight matrices might yield different results, though the negative controls suggest the mechanism itself is problematic. The experiment did not test varying coefficients for the entropy term, which might reveal a regime where the effect is neutral rather than negative. Finally, the subject-disjoint split tests generalization to new individuals but does not cover all forms of distribution shift, such as sensor drift or different activity definitions. Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.0125098 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

8. Conclusion

This study tested whether explicit entropy maximization of weight uncertainty parameters improves out-of-distribution accuracy on subject-disjoint time-series data. The results from both primary and confirmatory-replication cohorts refuted the hypothesis, showing that the intervention significantly worsened accuracy compared to standard deterministic training. The negative controls confirmed that the effect was specific to the entropy maximization objective. These findings change the recommendation for practitioners choosing between these methods: standard regularization should be preferred over entropy maximization for robustness in this setting. To settle the question further, future work should explore whether different entropy coefficients or architectural modifications can mitigate the observed performance degradation.

Appendix

Executed configuration. The reported run executed the calibrated testbed `uci_har_small_transformer_v1` exactly as registered.
- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- `uci_har_smartphones` — https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdbc81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

Seed identities. The primary and confirmatory-replication seed sets were fixed in the preregistration before any compute was spent, are disjoint, and every seed is included in the estimand.
- Primary (43): 497360171578835126, 619747419151856849, 4359311500859520721, 3900812323465083926, 1516445550462428226, 1630043136886007277, 1835401958502649079, 4224212089476219622, 2975544019134720152, 3507764674343915068, 4004663391355442268, 1509823930515868492, 3483967100743641674, 2225936250788509260, 1957754086644492888,

4154286735266072875, 3813494016017719200, 690472406820790188, 3813916678459859089, 4090465763900842853, 1844212296145829516, 3081923774287884327, 3477292413922411937, 726925812904865219, 4400454224498690158, 1309720931756809616, 3670000523134785215, 1572470746349086798, 5569990145947559, 2855297384752490100, 3166602034212987559, 4221459448664290905, 2934143650096726109, 1462463051419223747, 4363375170763905339, 3648570110071838823, 2581158873339944485, 2427086170940110629, 4084426303853737760, 2996901096067535381, 2815367292871095378, 4593572677630054665, 3754669615394449950.
- Confirmatory replication (43): 460845256865125083, 168360166801404394, 4330504984584448312, 2288886165882978077, 1436023229729468928, 3075137203164680509, 1264378101010998013, 2924740817759706416, 1407030776384369816, 3476452012804091485, 413443216560218742, 3744310097398416000, 611732904494353021, 3048141337216032385, 3348171016926346274, 182544617978886718, 2236541698554417841, 316787348038406038, 2542361651984418572, 2578784020613117302, 2827919878303280846, 50058912656477012, 2277661776842494339, 4538007595258168616, 2319057271103286511, 2045154589577734854, 2155395884799988780, 4226716155317983348, 1618031700484817618, 2260344274002574536, 2384851163006458576, 926629052757699259, 3894402378906575006, 299845134544061382, 460921128077554766, 3639277473403927291, 370580726517054298, 3492117404588741396, 3679724535907974316, 808192457140985599, 4061812732870876608, 820537500989758760, 2718212184032681684.

Reproducibility

All results were produced by an automated pipeline running the experiment on containerised GPU compute. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). The experiment ran over 43 preregistered primary seeds and 43 disjoint confirmatory-replication seeds; the seed identities are listed in the appendix. Compute job id: 15540577. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbc24249141b15ff622755. Walltime: 01:30:00. The exact code, container reference, seeds, run logs, job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: https://github.com/ABS-gmbh/ai-research-journal-papers.

Ethics

This is a small-scale computational study on publicly released data used under its published licence: uci_har_smartphones (CC BY 4.0). The data are publicly released, de-identified recordings of human volunteers collected and consented under the original data collection; no personal identifiers were accessed and no new data were collected from human subjects for this study. It poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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